Pattern Library v4

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Liebherr-IT Services

01
Pattern Library evolution

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01 Pattern Library evolution

v2

No more updates

v3

Accessibility

v4

Developer Experience

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02
Why is a new major release necessary

02 Why is a new major release necessary

Requirements

Support right to left, dark mode, global styles, performance (Outsourcing of assets: fonts, images and translations)

Refactoring

Extensive refactoring and redevelopment necessary

Standardisation

Standardisation of names for properties, events, etc.

Modern development stack

More open for new technologies

03
Strategy

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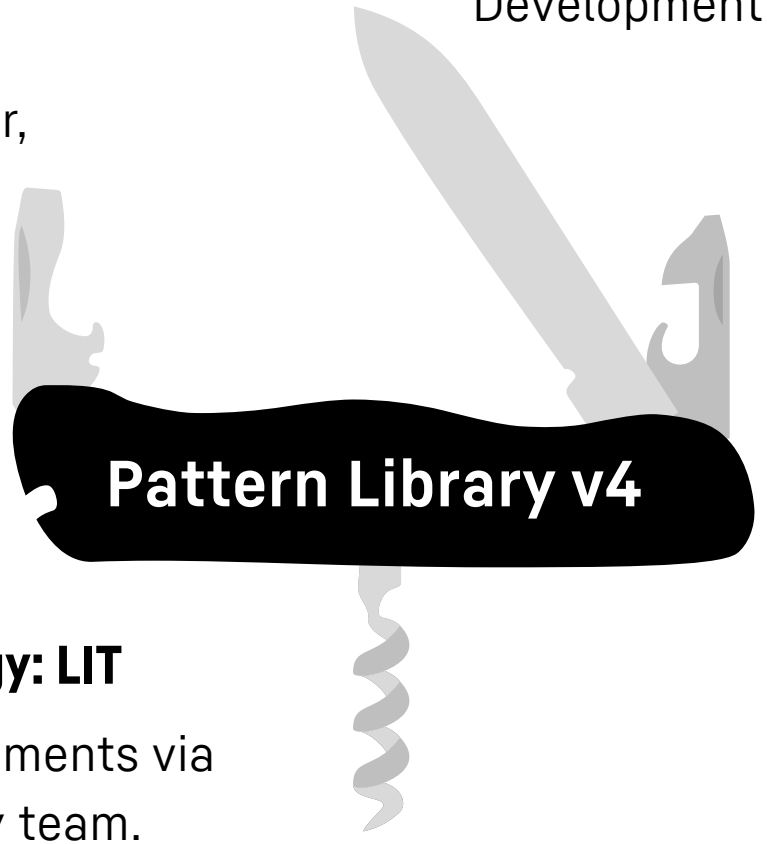
03 Strategy

Refactoring and new components

Development standards

Improved wrappers

Almost native behaviour in Angular, React, ...



Reduction of available components

"quality before quantity" (derived from usage statistics)

New technology: LIT

Faster developments via Pattern Library team.

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Benefits

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04 Benefits of new major release v4

Higher component quality

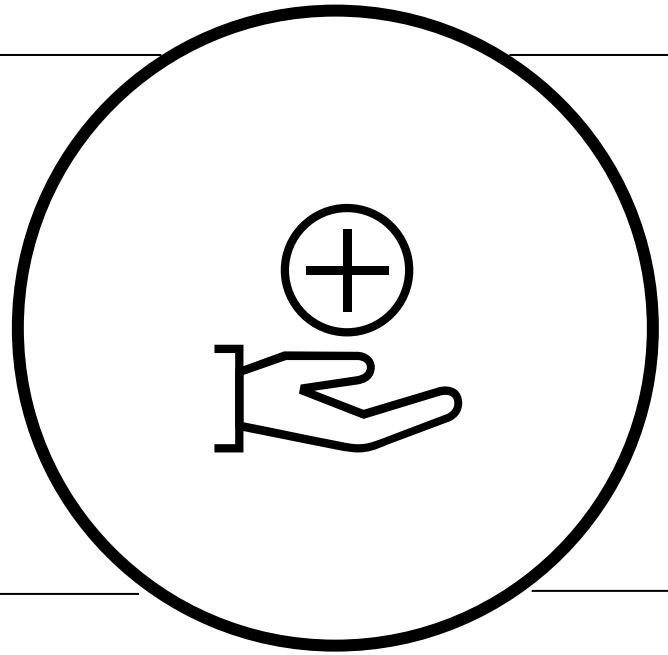
through reduction, redevelopment and refactoring

Better integration

V4 runs parallel with other versions

Better test coverage

Through new team member



Intuitive basic components

customisable via part selector

Improved documentation

with code examples for the individual frameworks

Better dev experience

New wrappers for a “native components” feel

05
Timeline

05 Timeline

11/25

12/25

01/25

Pre phase

- Create repo on Github
- Prepare JIRA
- POC: wrapper Angular, React,
- POC: SSR for NextJS
- Create example component with LIT
- Discuss, exchange with teams of LIS, LDC, LBC, LLI

Milestone v4.0
(Release planned on 2025-12-15)

- Button
- Icon
- AG Grid
- Tabs
- Loading spinner
- Link

Milestone v4.1

- Variables
- Overlay Container
- Concept form controls
- Headline
- Avatar
- Menu items
- Card
- Accordeon

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06
Icons in v4

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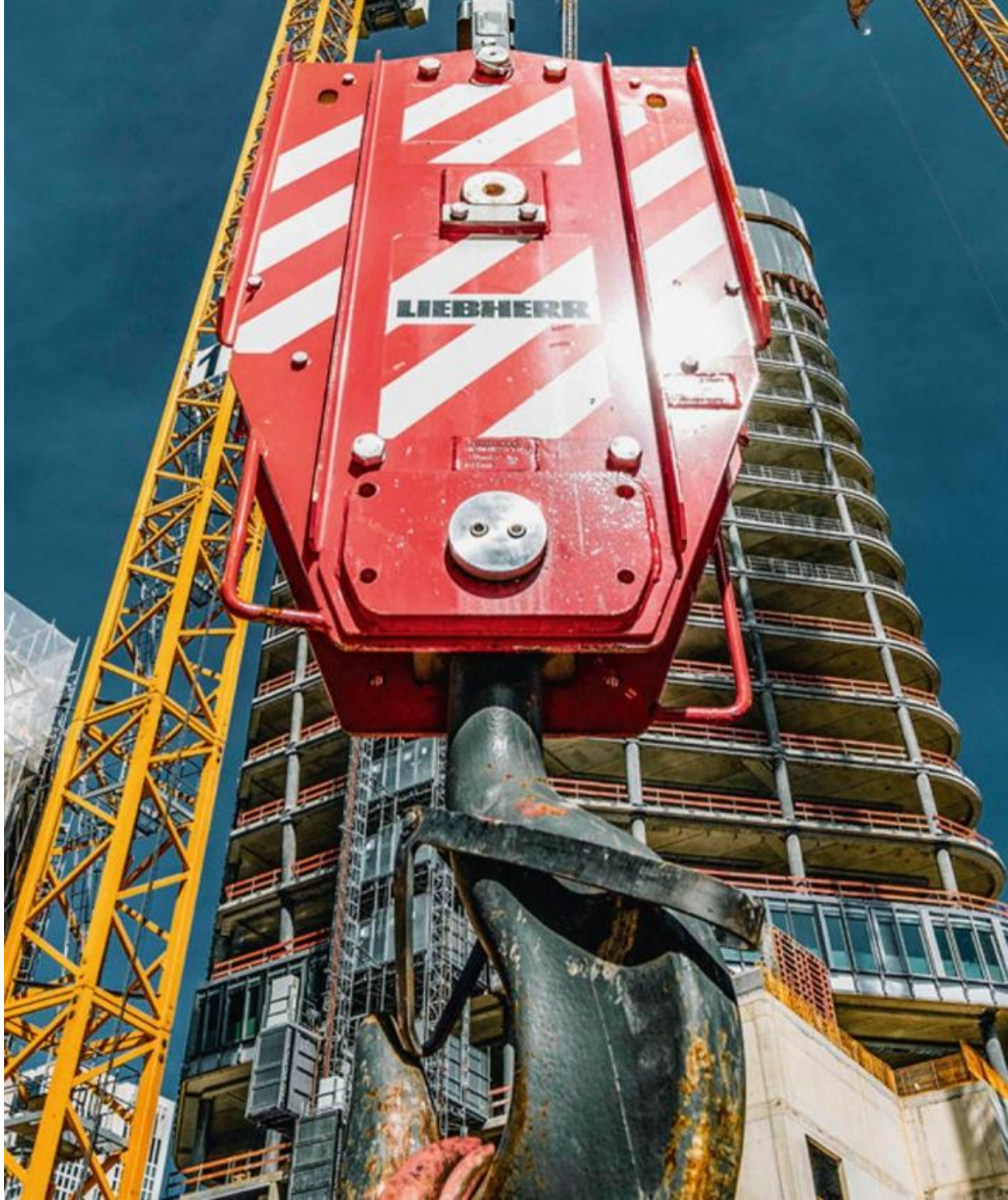
Icons in v4

- Situation
- Complications
- Question
- Answer

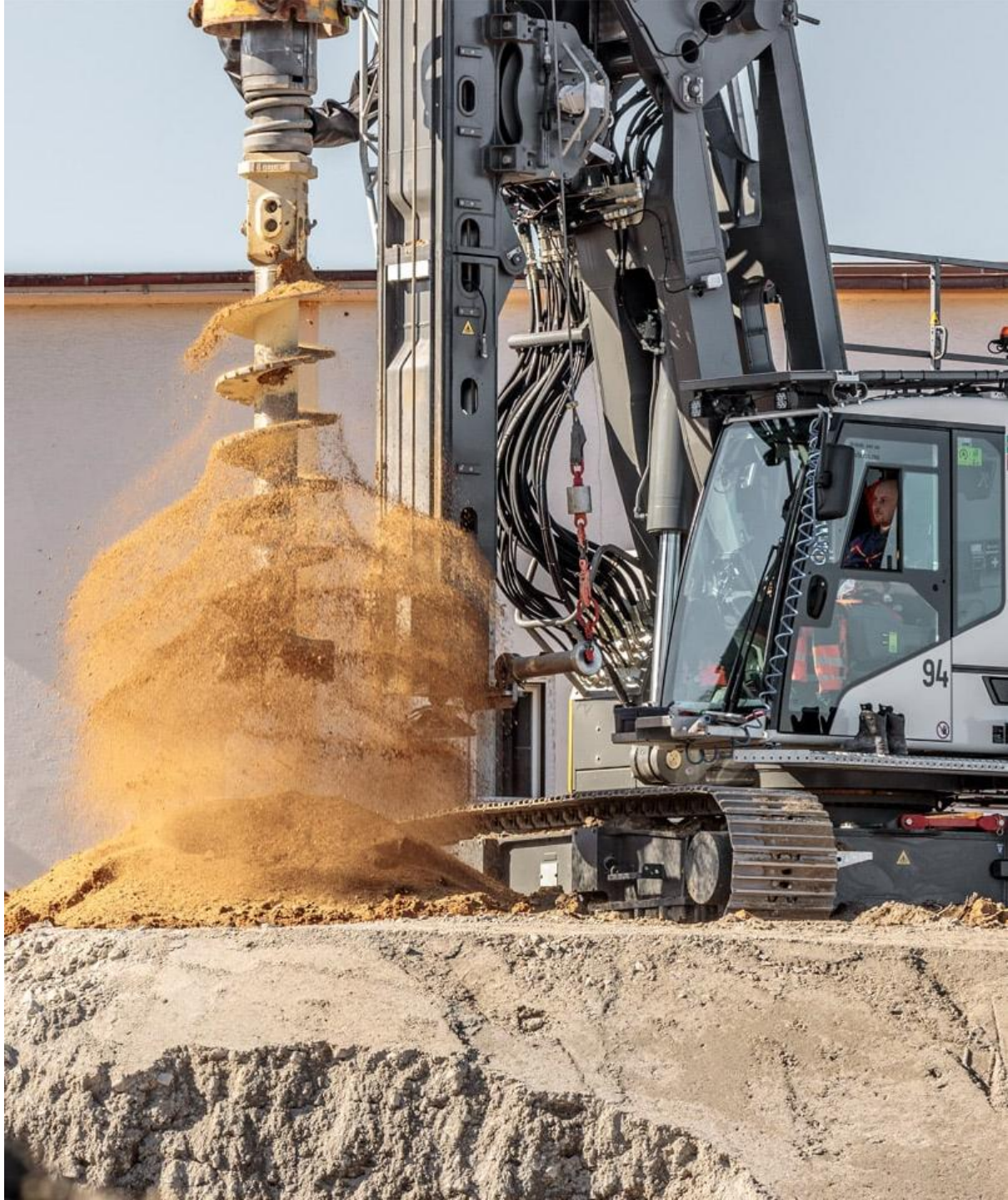


Situation

- 1200+ Icons
- Webfont
- Asset of Patternlib
- Icon Component

Complication

- Tightly coupled versioning
- Performance



Question

How to achieve looser coupling?

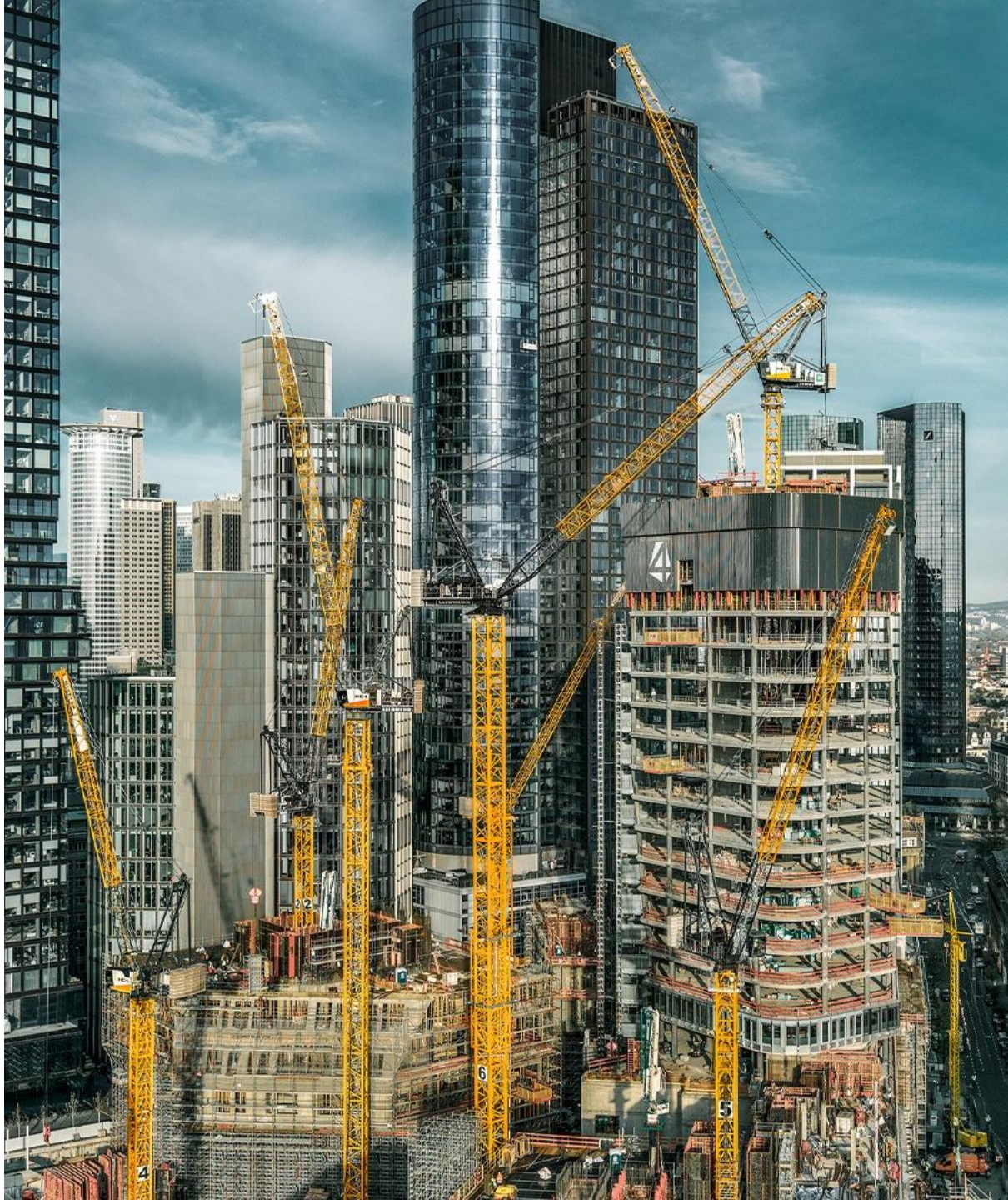
How to improve performance?

Answer

Separate packages

Enable tree shaking

Improved DevX



Separate Packages

Raw SVG, Vue, React, Angular

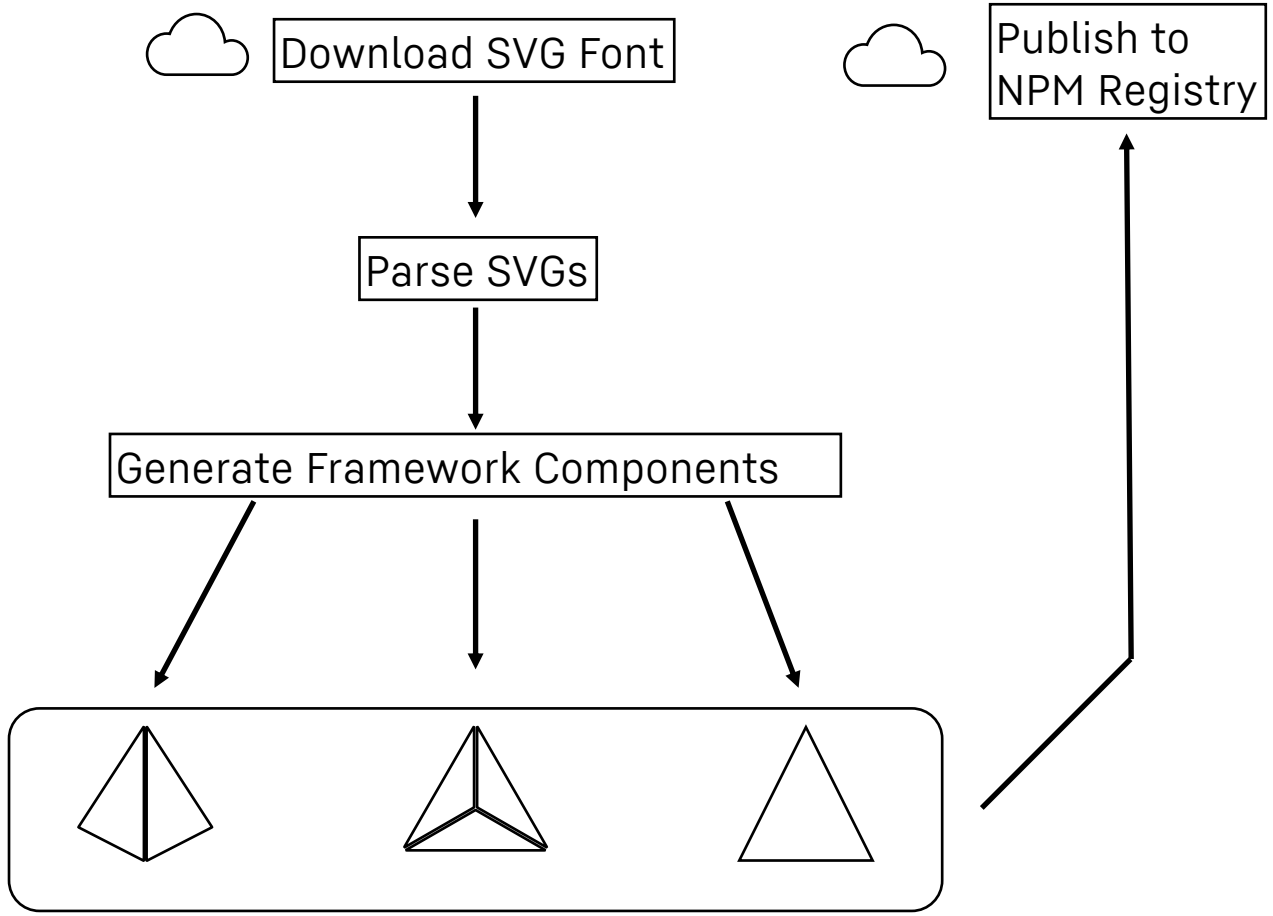
liebherr/icons-angular

liebherr/icons-react

liebherr/icons-vue

liebherr/icons-raw

How it works


Download SVG Font
Publish to NPM Registry
Parse SVGs
Generate Framework Components

HEXACON
IOI PROPERTIES

07
Part selector

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Problem

Pattern Library

Stable components

Intuitive components

Basic components

For general use

Combination is not a good solution – it leads to bugs, less intuitive components, and poor code quality.

Individual Teams

Special solutions

Custom design

More flexibility

Only suitable for their project

Solution

Part Selector

Patternlib provides basic components

+

You can customize components with part selector

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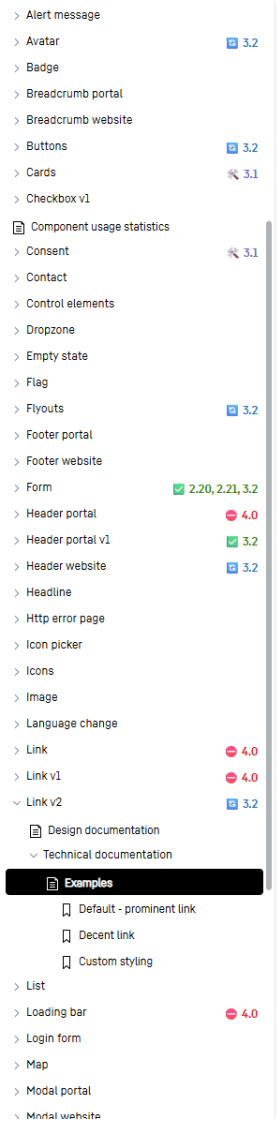
How?

Look up the parts

You can find the available **shadow parts** for each component in the documentation under the *Props* tab.



Examples | **Props** | Accessibility | Migration Guide

patternlib-link-v2

Properties

Property	Attribute	Description	Type	Default
active	active	Set link's active state.	boolean	false
ariaLabel	aria-label	Optional. Set the aria-label attribute.	string	undefined
colorInverted	color-inverted	Inverts colors for cases with dark background.	boolean	false
disabled	disabled	Set link's disabled state.	boolean	false
focusGapColor	focus-gap-color	Set the focus gap color.	"themeColor" | "transparent"	'transparent'
hideIcon	hide-icon	Hide icon, if not needed.	boolean	false
hovered	hovered	Set link's hovered state.	boolean	false
href	href	Set a hyperlink. (If not set, link will have an anchor but no focus)	string	''
icon	icon	Set an icon by icon name.	string	''
label	label	Set the Link label.	string	''
loading	loading	Show loading spinner.	boolean	false
noAnchorCase	no-anchor-case	For special cases where no anchor is needed	boolean	false
target	target	Define hyperlink's target. (If no underscore prefix is provided it will be added automatically)	"" | "_blank" | "_parent" | "_self" | "_top" | "blank" | "parent" | "self" | "top"	'blank'
type	type	Set link's type. (Prominent: Bold appearance, Decent: Light appearance)	"decent" | "prominent"	'prominent'
variant	variant	Set link's variant. (Action: Icon left, Context: Arrow icon right, Cross: Underlined)	"action" | "context" | "cross"	'action'

Shadow Parts

Part	Description
"patternlib-link-v2-label"	Targets the link's label.

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How?

Part name

A part's name is a combination of the component's name and the name of the element (part) within the component whose style you want to override.

patternlib-link-label

components name

part

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How?

Code example

```
.custom-class::part(patternlib-link-label) {
  font-size: 16px;
  text-transform: none;
}
```

1. Create your custom class
2. Use the part selector on it...
3. ...with the part name, you want to override

(inline css styling does not work)

```
<patternlib-link class="custom-class" label="My Link"></patternlib-link>
```

Use your just created class with the link

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How?

CSS
Pseudo classes / States

```
tabbed-custom-element::part(tab):hover {
        background-color: black; color: white;
}
```

Hover

```
tabbed-custom-element::part(tab active) {
        border-color: blue !important;
}
```

Active

JS
QuerySelector

```
document.querySelector("#tabbed-custom-element");
```

QuerySelector

More information

Link component example	*https://patternlibrary.liebherr.com/__prerelease/?path=/docs/components-link-v2-technical-documentation--examples*
Documentation part selector	*https://developer.mozilla.org/en-US/docs/Web/CSS/::part*

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08
Live demo

09
Questions

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Thank you!

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